

=== GNB (1785921472) ===

cu_cp:

amf:

```
addr: 10.53.1.2          # The address or hostname of the AMF.
port: 38412
bind_addr: 10.53.1.1     # A local IP that the gNB binds to for traffic from the AMF.
supported_tracking_areas:
- tac: 7
  plmn_list:
  - plmn: "00101"
    tai_slice_support_list:
    - sst: 1 # SST for eMBB (TS 23.501, Table 5.15.2.2-1)
inactivity_timer: 7200    # Sets the UE/PDU Session/DRB inactivity timer to 7200 seconds. Supported: [1 - 7200]
```

ru_sdr:

```
device_driver: zmq        # The RF driver name.
device_args: tx_port=tcp://127.0.0.1:2000,rx_port=tcp://127.0.0.1:2001,base_srate=11.52e6 # Optionally pass argument
srate: 11.52              # RF sample rate might need to be adjusted according to selected bandwidth.
tx_gain: 75               # Transmit gain of the RF might need to adjusted to the given situation.
rx_gain: 75               # Receive gain of the RF might need to adjusted to the given situation.
```

cell_cfg:

```
dl_arfcn: 368500         # ARFCN of the downlink carrier (center frequency).
band: 3                   # The NR band.
channel_bandwidth_MHz: 10 # Bandwidth in MHz. Number of PRBs will be automatically derived.
common_scs: 15            # Subcarrier spacing in kHz used for data.
plmn: "00101"            # PLMN broadcasted by the gNB.
tac: 7                    # Tracking area code (needs to match the core configuration).
pdccch:
  common:
    ss0_index: 0          # Set search space zero index to match srsUE capabilities
    coresets0_index: 6    # Set search CORESET Zero index to match srsUE capabilities
  dedicated:
    ss2_type: common      # Search Space type, has to be set to common
    dci_format_0_1_and_1_1: false # Set correct DCI format (fallback)
prach:
  prach_config_index: 1   # Sets PRACH config to match what is expected by srsUE
pdsch:
  mcs_table: qam64        # Sets PDSCH MCS to 64 QAM
pusch:
  mcs_table: qam64        # Sets PUSCH MCS to 64 QAM
```

log:

```
filename: /tmp/gnb.log    # Path of the log file.
all_level: info           # Logging level applied to all layers.
hex_max_size: 0
```

pcap:

```
mac_enable: false        # Set to true to enable MAC-layer PCAPs.
mac_filename: /tmp/gnb_mac.pcap # Path where the MAC PCAP is stored.
ngap_enable: true         # Set to true to enable NGAP PCAPs.
ngap_filename: /tmp/gnb_ngap.pcap # Path where the NGAP PCAP is stored.
e2ap_enable: true         # Set to true to enable E2AP PCAPs.
e2ap_du_filename: /tmp/gnb_du_e2ap.pcap # Path where the DU E2AP PCAP is stored.
e2ap_cu_cp_filename: /tmp/gnb_cu_cp_e2ap.pcap # Path where the CU-CP E2AP PCAP is stored.
e2ap_cu_up_filename: /tmp/gnb_cu_up_e2ap.pcap # Path where the CU-UP E2AP PCAP is stored.
```

#e2:

```
# enable_du_e2: true      # Enable DU E2 agent (one for each DU instance)
# e2sm_kpm_enabled: true  # Enable KPM service module
# e2sm_rc_enabled: true   # Enable RC service module
# addr: 127.0.0.1         # RIC IP address
# bind_addr: 127.0.0.100 # A local IP that the E2 agent binds to for traffic from the RIC. ONLY required if
# port: 36421             # RIC port
```

```
metrics:
  layers:
    enable_rlc: true          # Enable RLC metrics
    enable_sched: true       # Enable DU scheduler metrics
  periodicity:
    du_report_period: 400    # Set DU statistics report period to 400ms
```

=== UE (1785921472) ===

```
[rf]
freq_offset = 0
tx_gain = 50
rx_gain = 40
srate = 11.52e6
nof_antennas = 1

device_name = zmq
device_args = tx_port=tcp://127.0.0.1:2001,rx_port=tcp://127.0.0.1:2000,base_srate=11.52e6

[rat.eutra]
dl_eurfcn = 2850
nof_carriers = 0

[rat.nr]
bands = 3
nof_carriers = 1

max_nof_prb = 106
nof_prb = 52

[pcap]
enable = none
mac_filename = /tmp/ue_mac.pcap
mac_nr_filename = /tmp/ue_mac_nr.pcap
nas_filename = /tmp/ue_nas.pcap

[log]
all_level = info
phy_lib_level = none
all_hex_limit = 32
filename = /tmp/ue.log
file_max_size = -1

[usim]
mode = soft
algo = milenage
opc = 112233445566778899aabbccddeeff00
k = 00aabbccddeeff112233445566778899
imsi = 0010100000000001
imei = 353490069873319

[rrc]
release = 15
ue_category = 4

[nas]
apn = srsapn
apn_protocol = ipv4

[gw]
netns = ue1
ip_devname = tun_srsue
ip_netmask = 255.255.255.0

[gui]
enable = false
```

=== DOCKER (1785921472) ===

```
#
# Copyright 2021-2026 Software Radio Systems Limited
#
# This file is part of srsRAN
#
# srsRAN is free software: you can redistribute it and/or modify
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#
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# the LICENSE file in the top-level directory of this distribution
# and at http://www.gnu.org/licenses/.
#
```

services:

```
5gc:
  container_name: open5gs_5gc
  build:
    context: open5gs
    target: open5gs
    args:
      OS_VERSION: "22.04"
      OPEN5GS_VERSION: "v2.7.0"
  environment:
    - MONGODB_IP=127.0.0.1
    - OPEN5GS_IP=10.53.1.2
    - UE_IP_BASE=10.45.0
    - UPF_ADVERTISE_IP=10.53.1.2
    - DEBUG=false
    - SUBSCRIBER_DB=001010000000001,00aabbccddeeff112233445566778899,opc,112233445566778899aabbccddeeff00,8000,9,10.
    - NETWORK_NAME_FULL=srsRAN
    - NETWORK_NAME_SHORT=srsRAN
    - TZ=Europe/Madrid
  privileged: true
  ports:
    - "9898:9999/tcp"
  # Uncomment port to use the 5gc from outside the docker network
#   - "38412:38412/sctp"
#   - "2152:2152/udp"
  command: 5gc -c open5gs-5gc.yml
  healthcheck:
    test: [ "CMD-SHELL", "nc -z 127.0.0.20 7777" ]
    interval: 3s
    timeout: 1s
    retries: 60
  networks:
    ran:
      ipv4_address: ${OPEN5GS_IP:-10.53.1.2}

gnb:
  container_name: srsran_gnb
  # Build info
  image: srsran/gnb
  build:
    context: ..
    dockerfile: docker/Dockerfile
    args:
      OS_VERSION: "24.04"
```

```

environment:
  - GNB_IP=10.53.1.1
# privileged mode is required only for accessing usb devices
privileged: true
# Extra capabilities always required
cap_add:
  - SYS_NICE
  - CAP_SYS_PTRACE
volumes:
  # Access USB to use some SDRs
  - /dev/bus/usb/:/dev/bus/usb/
  # Sharing images between the host and the pod.
  # It's also possible to download the images inside the pod
  - /usr/share/uhd/images:/usr/share/uhd/images
  # Save logs and more into gnb-storage
  - gnb-storage:/tmp
# It creates a file/folder into /config_name inside the container
# Its content would be the value of the file used to create the config
configs:
  - gnb_config.yml
  - gnb_compose_config.yml
# Customize your desired network mode.
# current netowrk configuration creastes a private netwoek with both containers attached
# An alterantive would be `network: host`. That would expose your host network into the container. It's the easie
networks:
  ran:
    ipv4_address: ${GNB_IP:-10.53.1.3}
  metrics:
    ipv4_address: 172.19.1.3
# Start GNB container after 5gc is up and running
depends_on:
  5gc:
    condition: service_healthy
# Command to run into the final container
command: gnb -c /gnb_config.yml -c /gnb_compose_config.yml

```

configs:

gnb_config.yml:

```

content: |
  cu_cp:
    amf:
      addr: 10.53.1.2          # The address or hostname of the AMF.
      port: 38412
      bind_addr: 10.53.1.1    # A local IP that the gNB binds to for traffic from the AMF.
      supported_tracking_areas:
        - tac: 7
          plmn_list:
            - plmn: "00101"
              tai_slice_support_list:
                - sst: 1 # SST for eMBB (TS 23.501, Table 5.15.2.2-1)
      inactivity_timer: 7200 # Sets the UE/PDU Session/DRB inactivity timer to 7200 seconds. Supported: [

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ru_sdr:

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  common_scs: 15             # Subcarrier spacing in kHz used for data.
  plmn: "00101"              # PLMN broadcasted by the gNB.
  tac: 7                     # Tracking area code (needs to match the core configuration).
  pdcch:
    common:

```

```

    ss0_index: 0                # Set search space zero index to match srsUE capabilities
    cores0_index: 6             # Set search CORESET Zero index to match srsUE capabilities
dedicated:
    ss2_type: common            # Search Space type, has to be set to common
    dci_format_0_1_and_1_1: false # Set correct DCI format (fallback)
prach:
    prach_config_index: 1       # Sets PRACH config to match what is expected by srsUE
pdsch:
    mcs_table: qam64            # Sets PDSCH MCS to 64 QAM
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log:
    filename: /tmp/gnb.log      # Path of the log file.
    all_level: info             # Logging level applied to all layers.
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    mac_filename: /tmp/gnb_mac.pcap # Path where the MAC PCAP is stored.
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    e2ap_cu_cp_filename: /tmp/gnb_cu_cp_e2ap.pcap # Path where the CU-CP E2AP PCAP is stored.
    e2ap_cu_up_filename: /tmp/gnb_cu_up_e2ap.pcap # Path where the CU-UP E2AP PCAP is stored.

metrics:
    layers:
        enable_rlc: true        # Enable RLC metrics
        enable_sched: true      # Enable DU scheduler metrics
    periodicity:
        du_report_period: 400   # Set DU statistics report period to 400ms
gnb_compose_config.yml:
    content: |
        cu_cp:
            amf:
                addr: ${OPEN5GS_IP:-10.53.1.2}
                bind_addr: 0.0.0.0
        metrics:
            autostart_stdout_metrics: true
            enable_json: true
        remote_control:
            bind_addr: 0.0.0.0
            enabled: true

volumes:
    gnb-storage:

networks:
    ran:
        ipam:
            driver: default
            config:
                - subnet: 10.53.1.0/24
metrics:
    ipam:
        driver: default
        config:
            - subnet: 172.19.1.0/24

```

=== NOTAS (1785921472) ===

- PLMN (00101) y TAC (7) consistentes en gNB, UE y Core (SUBSCRIBER_DB).
- dl_arfcn (368500) para Banda 3 validado según REGLA 2.
- common_scs (15) utilizado.
- Puertos ZMQ (2000, 2001) cruzados entre gNB y UE según REGLA 4.
- bind_addr configurado según REGLA 5 (10.53.1.1 para gNB standalone, 0.0.0.0 para docker-compose).
- Bloque ru_sdr copiado exactamente del CAG según REGLA 6.
- IMSI, K, OPC y SUBSCRIBER_DB sincronizados y validados según REGLA 7.
- SST para eMBB (1) añadido con referencia a TS 23.501, Tabla 5.15.2.2-1.
- Métricas de RLC y scheduler DU habilitadas con periodo de 400ms.
- Capturas PCAP de NGAP habilitadas en gNB.
- IP estática del UE en SUBSCRIBER_DB ajustada a 10.45.0.2 para coincidir con UE_IP_BASE=10.45.0.
- nof_prb del UE ajustado a 52 para 10MHz de ancho de banda y 15kHz SCS.